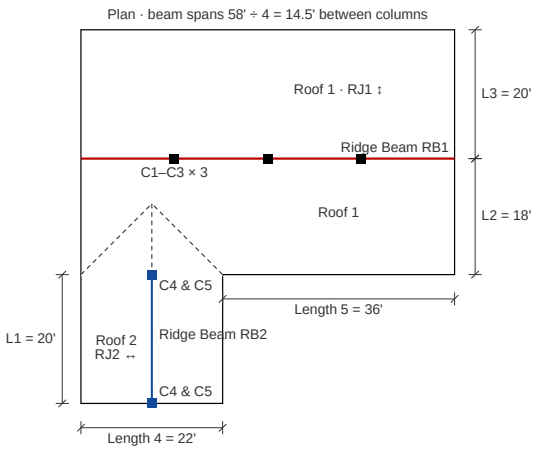
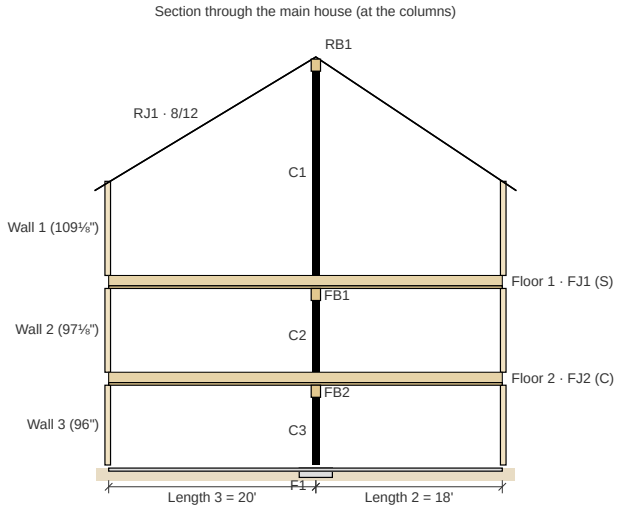
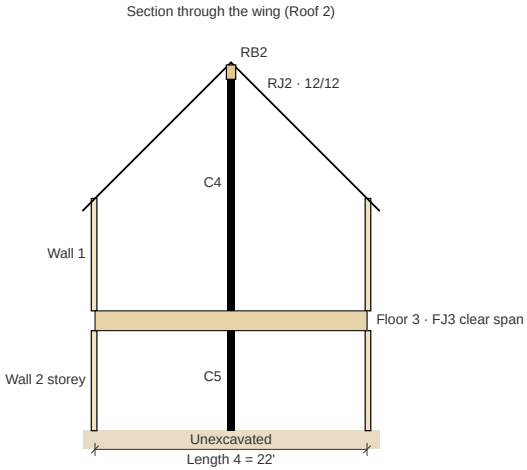
A5 – EWP – F24 (London) — worked key







Location

City	**London**

Roof 1

Pitch	**8**	/12
Overhang	**16**	inches
Dead load	**15**	psf
Number of interior columns	**3**	
RB1: Simple or continuous	**C**	S or C

Roof 2

Pitch	**12**	/12
Overhang	**12**	inches
Dead load	**15**	psf

Floor 1

Live Load	**40**	psf
Dead Load	**15**	psf
Joists: Simple or continuous	**S**	S or C
Directly applied ceiling	**Y**	Y or N
Beam: Simple or continuous	**C**	S or C

Floor 2

Live Load	**40**	psf
Dead Load	**15**	psf
Joists: Simple or continuous	**C**	S or C
Directly applied ceiling	**N**	Y or N
Beam: Simple or continuous	**C**	S or C

Floor 3

Live Load	**40**	psf
Dead Load	**25**	psf
Directly applied ceiling	**Y**	Y or N

General Information

Length 1	**20**	feet
Length 2	**18**	feet
Length 3	**20**	feet
Length 4	**22**	feet
Length 5	**36**	feet
Wall 1 height	**109.125**	inches
Wall 2 height	**97.125**	inches
Wall 3 height	**96**	inches
Subfloor thickness	**0.875**	inches
Soil capacity	**75**	KPa

Assumptions: 1. Walls have 0" width. 2. All beams and columns are weightless. 3. Footings sized on Total Factored Loads.

A5 – EWP – F24 (London) — worked key

Snow Loading

Ground snow load S_s	1.9	kPa
Associated rain load S_r	0.4	kPa
Specified snow load	1.445	kPa

0.55 × 1.9 + 0.4 = **1.445 kPa**

Specified snow load	30.20	psf

1.445 × 20.9 = **30.20 psf**. His key keeps **2 decimals** here.

Roof Joists RJ1

Unfactored snow load	30.20	psf
Unfactored dead load	15	psf
Pitch	8	/12
Maximum span	20	feet

Walls are 0" wide, so the span is the length to the ridge — the longer side, Length 3: **20'**

Overhang	16	inches
Joist depth	16"	inches

S = 30.20 psf → **40LL + 15DL** column (over 30.00, so not the 30 column). 8/12 is **High** slope (Low ≤ 6/12). Need ≥ 20' = 20'-0". Any spacing/depth/series in that column whose span is long enough is right.

Joist type	s33	
Spacing	24"	inches
Slope length	307.67	inches

(20 × 12 + 16) × √(1 + (8/12)²) = 256 × 1.2019 = **307.67"** (p.21 slope factor 1.202 gives the same within ½")

D factor	10.67	inches

16 × 8 ÷ 12 = **10.67"** (his key). Table p.21 says 10¾" — both accepted.

Length to order	27	feet

(307.67 + 10.67) ÷ 12 = 26.53' → round up = **27'**

Floor Joist FJ1

Unfactored live load	40	psf
Unfactored dead load	15	psf
Subfloor thickness	0.875	inches
Directly applied ceiling	Y	Y or N
Simple or continuous span	S	S or C
Span	20	feet

Walls 0" wide, beam on the line: spans are 20' and 18'. Size for the longer one: **20'**

Joist depth	11⅞"	inches

p.8: 40 LL + 15 DL, 7/8" OSB, directly applied ceiling, "simple or continuous" columns. Need ≥ 20'-0".

Joist type	s47	
Spacing	19.2"	inches
Length to order	20	feet

walls 0" wide: the span itself, rounded up to a whole foot = **20'**

Floor Joist FJ2

Unfactored live load	40	psf
Unfactored dead load	15	psf
Subfloor thickness	0.875	inches
Directly applied ceiling	N	Y or N
Simple or continuous span	C	S or C
Span	20	feet

Walls 0" wide, beam on the line: spans are 20' and 18'. Size for the longer one: **20'**

Joist depth	11⅞"	inches

p.8: 40 LL + 15 DL, 7/8" OSB, no directly applied ceiling, "continuous" (either table works — continuous-only spans are longer) columns. Need ≥ 20'-0".

Joist type	s47	
Spacing	16"	inches
Length to order	38	feet

one continuous joist over the beam: 20 + 18 = **38'**

Roof Joists RJ2

Unfactored snow load	30.20	psf
Unfactored dead load	15	psf
Pitch	12	/12
Maximum span	11	feet

Walls are 0" wide, so the span is the length to the ridge — half of Length 4: **11'**

Overhang	12	inches
Joist depth	16"	inches

S = 30.20 psf → **40LL + 15DL** column (over 30.00, so not the 30 column). 12/12 is **High** slope (Low ≤ 6/12). Need ≥ 11' = 11'-0". Any spacing/depth/series in that column whose span is long enough is right.

Joist type	s31	
Spacing	24"	inches
Slope length	203.65	inches

(11 × 12 + 12) × √(1 + (12/12)²) = 144 × 1.4142 = **203.65"** (p.21 slope factor 1.414 gives the same within ½")

D factor	16	inches

16 × 12 ÷ 12 = **16"** (his key). Table p.21 says 16" — both accepted.

Length to order	19	feet

(203.65 + 16) ÷ 12 = 18.3' → round up = **19'**

Floor Joist FJ3

Unfactored live load	40	psf
Unfactored dead load	25	psf
Subfloor thickness	0.875	inches
Directly applied ceiling	Y	Y or N
Span	22	feet

Clear span wall to wall (Length 4): **22'**

Joist depth	11⅞"	inches

p.8: 40 LL + 25 DL, 7/8" OSB, directly applied ceiling, "simple or continuous" columns. Need ≥ 22'-0".

Joist type	s47	
Spacing	12"	inches
Length to order	22	feet

walls 0" wide: the span itself, rounded up to a whole foot = **22'**

Ridge Beam RB1

Span	14.50	feet

(Length 4 + Length 5) ÷ (3 columns + 1) = 58 ÷ 4 = **14.5'**

Tributary width	19	feet

(20 + 18) ÷ 2 = **19'** — half of each joist span lands on the beam.

Unfactored snow load	574	plf

30.20 × 19 = 573.8 → round up = **574**

Unfactored dead load	285	plf

15 × 19 = 285 → round up = **285**

Unfactored total load	859	plf

574 + 285 = **859**

Total factored load	1,218	plf

1.5 × 574 + 1.25 × 285 = 1217.25 → round up = **1,218**

Beam choice	LVL 5¼×14	

Use the **16.5'** row (round 14.5' up). Need LL ≥ 574, TL ≥ 859, TFR ≥ 1,218. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height	14	inches
Bearing requirements	6.4	inches

Bottom row of the table for your beam: **2.6" end / 6.4" intermediate**. Beam is continuous over the columns → intermediate is the one that matters; both are accepted.

Floor Beam FB1

Span	14.50	feet

(Length 4 + Length 5) ÷ (3 columns + 1) = 58 ÷ 4 = **14.5'**

Tributary width	19	feet

(20 + 18) ÷ 2 = **19'** — half of each joist span lands on the beam.

FJ1: Simple or continuous	S	S or C
Multiplier for S or C span	1	

Simple joists: **1**

Unfactored live load	760	plf

40 × 19 = 760 → round up = **760**

Unfactored dead load	285	plf

15 × 19 = 285 → round up = **285**

Unfactored total load	1,045	plf

760 + 285 = **1,045**

Total factored load	1,497	plf

1.5 × 760 + 1.25 × 285 = 1496.25 → round up = **1,497**

Beam choice	PSL 7×14	

Use the **16.5'** row (round 14.5' up). Need LL ≥ 760, TL ≥ 1,045, TFR ≥ 1,497. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height	14	inches
Bearing requirements	7.2	inches

Bottom row of the table for your beam: **2.9" end / 7.2" intermediate**. Beam is continuous over the columns → intermediate is the one that matters; both are accepted.

Floor Beam FB2

Span **14.50** feet

(Length 4 + Length 5) ÷ (3 columns + 1) = 58 ÷ 4 = **14.5'**

Tributary width **19** feet

(20 + 18) ÷ 2 = **19'** — half of each joist span lands on the beam.

FJ2: Simple or continuous **C** S or C

Multiplier for S or C span **1.25**

Continuous joists put 25% more on the beam: **1.25**

Unfactored live load **950** plf

40 × 19 × 1.25 = 950 → round up = **950**

Unfactored dead load **357** plf

15 × 19 × 1.25 = 356.25 → round up = **357**

Unfactored total load **1,307** plf

950 + 357 = **1,307**

Total factored load **1,872** plf

1.5 × 950 + 1.25 × 357 = 1871.25 → round up = **1,872**

Beam choice **PSL 7×14**

Use the **16.5'** row (round 14.5' up). Need LL ≥ 950, TL ≥ 1,307, TFR ≥ 1,872. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height **14** inches

Bearing requirements **7.2** inches

Bottom row of the table for your beam: **2.9" end / 7.2" intermediate**. Beam is continuous over the columns → intermediate is the one that matters; both are accepted.

Ridge Beam RB2

Span **20** feet

Length 1 (wing): **20'**

Tributary width **11** feet

Length 4 ÷ 2 = 22 ÷ 2 = **11'**

Unfactored snow load **333** plf

30.20 × 11 = 332.2 → round up = **333**

Unfactored dead load **165** plf

15 × 11 = 165 → round up = **165**

Unfactored total load **498** plf

333 + 165 = **498**

Total factored load **706** plf

1.5 × 333 + 1.25 × 165 = 705.75 → round up = **706**

Beam choice **LVL 5¼×14**

Use the **20'** row (round 20' up). Need LL ≥ 333, TL ≥ 498, TFR ≥ 706. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height **14** inches

Bearing requirements **2.1** inches

Bottom row of the table for your beam: **2.1" end / 5.3" intermediate**. Simple beam → end is the one that matters; both are accepted.

Interior Column C1

RB1: Simple or continuous **C** S or C

Multiplier for S or C span **1.25**

Continuous beam puts 25% more on the column: **1.25**

Total factored load **22,077** pounds

RB1 TFL 1,218 × span 14.5' × 1.25 = 22076.25 → round up = **22,077 lb**

Column height **19.93** feet

rise 18 × 8 (= 18' × 8/12 × 12 in) + Wall 1 109⅛" − RB1 14" = 239.125" ÷ 12 = **19.93'** — his key takes the rise over **Length 2**.

Column choice **PSL 7×7**

Round 19.93' up to the next row; need ≥ 22,077 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Interior Column C2

FB1: Simple or continuous **C** S or C

Multiplier for S or C span **1.25**

Continuous beam puts 25% more on the column: **1.25**

Total factored load **49,211** pounds

C1 22,077 + FB1 1,497 × 14.5 × 1.25 (27,134) = **49,211 lb** — C2 holds up C1 too.

Column height **6.93** feet

Wall 2 97⅛" − FB1 14" = 83.125" ÷ 12 = **6.93'**

Column choice **PSL 5¼×5¼**

Round 6.93' up to the next row; need ≥ 49,211 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Interior Column C3

FB2: Simple or continuous **C** S or C

Multiplier for S or C span **1.25**

Continuous beam puts 25% more on the column: **1.25**

Total factored load **83,141** pounds

C2 49,211 + FB2 1,872 × 14.5 × 1.25 (33,930) = **83,141 lb** — everything above lands here.

Column height **6.83** feet

Wall 3 96" − FB2 14" = 82" ÷ 12 = **6.83'**

Column choice **PSL 7×7**

Round 6.83' up to the next row; need ≥ 83,141 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Column C4

Total factored load **7,060** pounds

RB2 TFL 706 × 20' ÷ 2 (simple beam, one end) = **7,060 lb**

Column height **18.93** feet

rise 11 × 12 + Wall 1 109⅛" − RB2 14" = 227.125" ÷ 12 = **18.93'**

Column choice **PSL 5¼×5¼**

Round 18.93' up to the next row; need ≥ 7,060 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Column C5

Total factored load **7,060** pounds

Same as C4: **7,060 lb**. FJ3 clear-spans wall to wall, so Floor 3 adds nothing to C5.

Column height **8.09** feet

storey below Floor 3: Wall 2 97⅛" ÷ 12 = **8.09'** (unconfirmed — ask in class) **(ask in class)**

Column choice **LSL 3½×3½**

Round 8.09' up to the next row; need ≥ 7,060 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Column Footing F1

Total factored load on C3 **83,141** pounds

Soil capacity **75** KPa

Soil capacity **1567.5** psf

75 × 20.9 = **1567.5 psf**

Footing area **53.04** square

83,141 ÷ 1567.5 = **53.04 ft²**

Length of square footing **87.39** inches

√53.04 = 7.283' × 12 = **87.4"**

Joist, beam and column boxes have more than one right answer — the key shows one that works. Loads use ROUNDUP like his A5 key; snow psf kept to 2 decimals.